

Practica 9:

Certificados Digitales

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Martes, 12 de mayo, 2026

Código:

```
from Crypto.PublicKey import RSA
from Crypto.Signature import pss
from Crypto.Hash import SHA3_256

# Función para Generar Llaves
def generar_llaves():
    print("\nGenerando par de claves RSA...")
    key = RSA.generate(2048)

    kpriv = key.export_key('PEM')
    kpub = key.publickey().export_key('PEM')

    return kpriv, kpub

# Función para generar certificado
def generar_certificado(kpriv_firmante_pem, kpub_sujeto_pem):
    print("\nGenerando certificado X.509...")

    datos_cert = {
        "CN": "Areli Guevara",
        "O": "IPN ESCOM",
        "EXP": "2026-05-12",
        "PUB": kpub_sujeto_pem.decode('utf-8')
    }

    m = SHA3_256.new(str(datos_cert).encode('utf-8'))
    key_obj = RSA.import_key(kpriv_firmante_pem)
    firmador = pss.new(key_obj)
    firma = firmador.sign(m)

    with open("kpriv_rsa.pem", "wb") as f:
        f.write(kpriv_firmante_pem)

    with open("certificado_rsa.pem", "w") as f:
        f.write(f"DATOS DEL CERTIFICADO\n{datos_cert}\n")
        f.write(f"FIRMA RSASSA-PSS\n{firma.hex()}")
```

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print("Certificado RSA y clave privada guardados con éxito.")

if __name__ == "__main__":
    kpriv_f, kpud_f = generar_llaves()
    kpriv_s, kpud_s = generar_llaves()

    generar_certificado(kpriv_f, kpud_s)

```

Ejecución:

```

1  DATOS DEL CERTIFICADO
2  {'CN': 'Areli Guevara', 'O': 'IPN ESCOM', 'EXP': '2026-05-12', 'PUB':
   '-----BEGIN PUBLIC
   KEY-----\nMIIBIjANBgkqhkiG9w0BAQEFAAOCAQIAMIIBcGKCAQEAXDHF1ZDOYvZ1HLSJL
   wWU/nkayJrO
   +lSMIPBx1fCSACnnHNo8RLUgZR0VYCKtarMwPdG8V4rvFkkEuPdMjH54g\n41MPcw5MY0w
   V7HiF1
   +xBWmk9j3qHsAHG1E9B3HyjNbmOpkxmFIZHxQVt1ckRN\nDwLMKYonEYys70Q10uTio
   55NsemwEz45mFXQZ5Eaj0LT5Za9j0DjvcnNwHqkRA\nKKZsbMcKGM3zswxONLio1Te05d2
   kUuHQf+zrr+cpDtGgjp8ifftrHSE40aDRm8Zf\nN9ZemKoQX
   +UXypMLlMx51JirDBbjR5zoak3ur/
   oLxrcT50EHVAsBQlMyuYuidsn\ntwIDAQAB\n-----END PUBLIC KEY-----'}
3  FIRMA  RSASSA-PSS
4  2acd11d15840eb446f9ebb11eae1ed197a2e3f1097552317a3326ae8653815361737053
   1adb881aedc39e2d3236f329bd98d502b75eef4ef622dde2c507ee148d93491d9a07a40
   f8e838759efb44dc97da24a477cc715dc93b243fea88257154ef307754e49bb22980f30
   d4df51f26f9a56c5ca2fec181d094ccae96a3273d8c2787edd8dbd69d2785e50cfe87
   b90b52e64015ddb5ee42336e959624aa6d227e59974e8095b458827f3fca0e9ada4faea
   6eb5c91d34019de0dca75c2d3b877d55df0063f092063e63c231d16388212d96c6283
   015b4cBecc13e8b1f36036ed3e19d408aae89710bccfd70b4b1b383f6218ce081514dd8
   5d070a9f99ddeb

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```

1  -----BEGIN RSA PRIVATE KEY-----
2  MIIEpAIBAAKCAQEAAnB+XYRTtm6kYDn5YI2QmhuCUV4wY9gwiMwIgzSb0gg6mDST
3  XVLKU33dSbog109GMY1etkyPnQA4na3znmYtiaGevC1Spqddj6zfb2PLqSa7q2W
4  +M/NxrgmktDr0PDZ7RDIZwx3q9FbGRYUahPV+7iUi+7ieMfesecmznNmom1ZX2vS
5  4G0J1ud18o8+7RcyHgEGgf+9Zjh5XxkMEXyLfnudYYQuRghPhdLBU0xkiHApCw3
6  3/T/VAGM922o2ZkGwIP2BAMZbccaArAzxt+iS92aHkr0xrBDU0U6SkGg+xUGB41b
7  goX0QgHofhv1ikkr2gtrsrhpyNGUaiWJPoolfQIDAQABAOIBABS1GfP98KAPVw0B
8  GnL4PKXyHqrYAa2M4XdyjQA47kdb4Sez/YHJqJkXZKT1Q4r-bUGedJR1P3VKJJ3k+
9  i8mmFoblQV+HCqWvp3UTn7M4UPVw0YFjzs0MMp1PWSHnxz1/Va0WRrC5ficnZUF
10 vH15DQ5EpqtCzYViICXNK34X30witGCE7HFeyeoA0wAvjju/ookiF01NCm0/7TA
11 wkbyI25n0SFus+r4wf8wnv9n6VTSQJxMAQXJC1hwQ4ppf7GkDhdZpGJrwOKOMgc
12 yQw8Ena2rXmoqSTDE09TAHmc7TmKYDwQL1uW9vEpYp+ppDpbEELUN3Y5Ik/XCinkH
13 zkV/5L8CgYEAwt15Chf3V/1zb1Mx1Uw080j111e0Xqk3nRlryK5kx7g2PACxaf+n
14 ortdt2i7TqkXRVUTkN54GQcxDA/QGSZJ1cqdvUplrt1rweZf939HLsQB0tdfozg
15 H7dcEKLNBdpXGarqxrokXs/stDpCcrdiDcv92L4Hkf+mnsCb0fufB1sCgYEAzRM
16 7EBL7EclHz068sk9Vxc2fKx6h4JG6tutYFeQDh4ndJ670dt/Y1VugcXVuzN1q17
17 J0bps6yJdmvvehg83Yxtgf/w0trdefqQdUp4dsLRGMKSXZ6KF0vqFy9dmJAQu8Q
18 uDZR617P/JZNwY3Jx+S60H0wGfUzt/9rc9oiAcGgYAw0ubopj1fQGLIUmtCOX5
19 08wTVzP4V+56Jnabwc76moB1ldG8aRm3gCVCzSImYf7lqYfqr20Vi/JK5Z5idaQs
20 p1rNTGKYovYJAcqPBy0M1VjM3wcZhg1cEGjr-fUDX20d7kQ04G88wMq4ectE1fuh
21 aqL06WS6UMuPdCfTPprZDQKBgQGBMoH+Tpsa4xm0B6seKIDBAHR21GaEqj03n/j
22 11R2JBR3J88OWILRmh+ogb30bNS8ddXAFjve/N3nu00D1tgWNA8yM2y11Xq0ZKM9
23 0q+Qhoq0BrftQ7dCYfg/UbZC8SMMWJms+n5s5p0Wjv0czt8p9izp9u9mCyzDLC
24 mxq2QK8GQDCANDs3JkBoPGKdCazq0XFKM0WkdpnIBF9JUmMDNprc8BmAXozhsJ

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```

9/src/p9.py"

Generando par de claves RSA...

Generando par de claves RSA...

Generando certificado X.509...
Certificado RSA y clave privada guardados con éxito.

```